

CURRENT & RESISTANCE

Q.1 What is electric current and current density, write its units?

ELECTRIC CURRENT

The rate of flow of charges through a conductor is called electric current. The electric current I is established in a conductor when charge dq passes through a cross section of a conductor in time dt

$$I = \frac{dq}{dt}$$

UNIT

The SI unit of electric current is called ampere. The current established in conductor is equal to one ampere when one coulomb charge flows through a conductor in one second.

CHARGE CARRIERS

The charge carries in metallic conductors are free electrons.

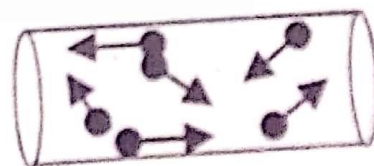
The charge carriers are positive and negative ions in electrolytic conductors.

The charge carriers are electrons or ions in gases.

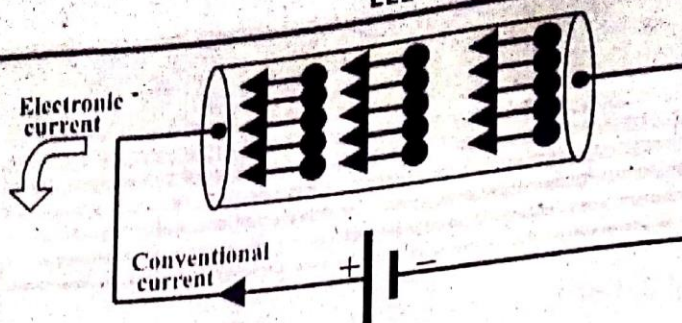
DIRECTION OF CURRENT

The free electrons are in continuous random motion in metallic conductors. The current is not established without external electric field whether conductor is charged or uncharged.

When external electric field is applied, electrons move from low potential terminal(-ve) to high potential terminal(+ve) of battery. The current due to movement of electrons is called electronic current.



The direction of conventional current in a circuit is from high potential to low potential terminal of battery.



CURRENT DENSITY

The current per unit cross sectional area of a conductor is called current density. It is denoted by $\vec{J}$. It is a vector quantity. Its direction is same as applied electric field. Its SI unit is called A/m^2 .

$$J = \frac{dI}{da}$$

$$I = \int \vec{J} \cdot d\vec{a}$$

The above eq becomes when J is constant over the surface of conductor and right angle to cross section area A .

$$I = J A$$

$$J = \frac{I}{A}$$

The direction of current density $\vec{J}$ and applied electric field $\vec{E}$ is same but direction of drift velocity $\vec{V}_d$ of electrons is opposite to direction of current density.

Q.2 What is drift velocity. Establish a relation between drift velocity and current density?

DRIFT VELOCITY

The free electrons move from low potential terminal to high potential terminal of battery when a conductor is connected with a battery. These free electrons collide with ionic cores of the conductor during motion. The kinetic energy of accelerating electrons is converted into vibrational energy of the lattice.

The constant average velocity acquired during this process by free electrons against electric field is called drift velocity V_d . The drift velocity is of the order 10^{-3} m/s. The random thermal motion of conduction electrons in a metal takes place with typical speeds of 10^6 m/s.

RELATION BETWEEN DRIFT VELOCITY AND CURRENT DENSITY

Consider a conductor having length L and cross sectional area A . The volume of conductor is AL .

TOTAL CHARGE CARRIERS

Unit volume of conductor has charge carriers = n

AL volume of conductor has charge carriers = nAL

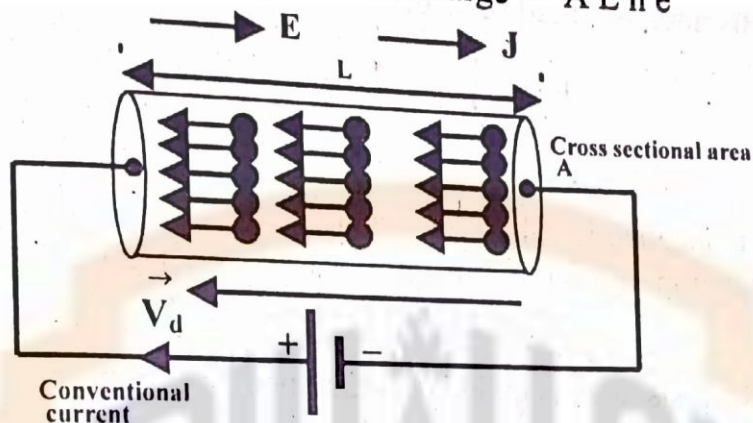
Total charge carriers in conductor = nAL

TOTAL CHARGE

Now consider a single charge carrier in conductor

One charge carrier has charge = e

nAL charge carriers have charge = $ALne$



The total charge q on conductor is

$$q = nALe \quad \text{-----(1)}$$

The current established in conductor when charge q flows across the conductor in time t is

$$I = \frac{q}{t}$$

$$q = It \quad \text{-----(2)}$$

CURRENT DENSITY

Comparing eq(1) and eq(2)

$$I \times t = nALe$$

The charges moves with drift velocity V_d across the conductor having length L in time t so $L = V_d t$

$$I \times t = nA(V_d t)e$$

$$I = nAV_d e$$

$$\frac{I}{A} = neV_d$$

$$J = neV_d$$

FORMULA OF DRIFT VELOCITY

$$V_d = \frac{J}{ne}$$

This is relation between current density and drift velocity of charge carriers in conductor. The direction of drift velocity and current density is opposite to each other, so

$$\vec{J} = -ne\vec{V_d}$$

we can write this in vector form as

Q. 3 Define electrical resistance, conductance, resistivity, conductivity and write their units?

ELECTRICAL RESISTANCE

The measure of opposition to flow of charges through a conductor due to collision with ionic cores of conductor is called electrical resistance.

Mathematically the ratio of pd applied at the ends of a conductor and current flowing through conductor is equal to resistance. The conductor used in a circuit to provide specified resistance is called resistor.

$$R = \frac{V}{I}$$

UNIT

The SI unit of resistance is called ohm (Ω). The resistance is called one ohm when one volt pd causes one ampere flow of current through a conductor.

CONDUCTANCE

The reciprocal of resistance is called conductance. It is denoted by G.

$$G = \frac{1}{R} = \frac{I}{V}$$

UNIT

The SI unit of conductance is called Ω^{-1} or siemen.

RESISTIVITY

The ratio of electric field E applied to a conductor and current density J is called resistivity. It is denoted by ρ . The resistance of one (meter)³ of a material is called resistivity. The resistivity of copper is $1.7 \times 10^{-7} \Omega - m$. The resistivity of a conductor depends upon temperature.

$$\rho = \frac{E}{J}$$

UNIT

The SI unit of resistivity is called ohm – meter ($\Omega - m$)

CONDUCTIVITY

The reciprocal of resistivity is called conductivity. It is denoted by σ .

$$\sigma = \frac{1}{\rho} = \frac{J}{E}$$

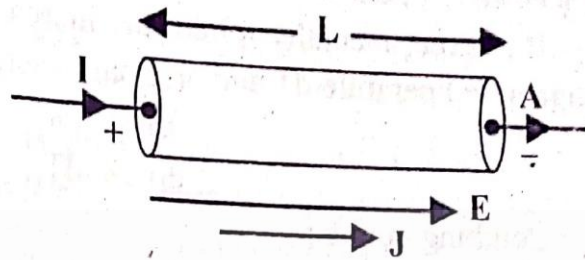
UNIT

The SI unit of conductivity is called $(\Omega - m)^{-1}$. The unit mho per meter is also used for measurement of conductivity.

Q. 4 A potential V is applied across length L of a conductor. Relate (i) J with ρ and E (ii) E with V and L . Hence prove that $\rho = RA/L$? 155

RELATION BETWEEN RESISTANCE AND RESISTIVITY

Consider a cylindrical conductor having length L and cross sectional area A . The current I flows through conductor when pd V is applied at the ends of a conductor. The electric field E and pd V are related as



$$E = \frac{V}{L}$$

$$E = \frac{IR}{L} \quad \text{----- (1)}$$

The resistivity of the material is given as

$$\rho = \frac{E}{J}$$

$$E = \rho J \quad \text{----- (2)}$$

Comparing eq(1) and eq(2)

$$\rho J = \frac{IR}{L}$$

$$J = \frac{IR}{\rho L} \quad \text{----- (3)}$$

The current density J is defined as

$$J = \frac{I}{A} \quad \text{----- (4)}$$

Comparing eq(3) and eq(4)

$$\frac{I}{A} = \frac{IR}{\rho L}$$

$$R = \frac{\rho L}{A}$$

$$\rho = \frac{RA}{L}$$

This is relation between resistivity and resistance. This relation shows that resistance of a conductor is directly proportional to length of conductor and inversely proportional to cross sectional area of conductor.

Q. 5 What is temperature coefficient of resistivity. Derive its formula and explain effect of temperature variation on resistivity?

TEMPERATURE COEFFICIENT OF RESISTIVITY

The fractional change in resistivity per unit change in temperature is called temperature coefficient of resistivity. It is denoted by α . Its SI unit is called $(\text{kelvin})^{-1}$.

FORMULA

Consider a conductor having resistivity ρ_0 at temperature T_1 . The resistivity of conductor increases by amount $d\rho = \rho_t - \rho_0$ due to increase in vibrational motion of ionic cores when temperature of conductor is increased through $dT = T_2 - T_1$.

It is experimentally found that increase in resistivity $d\rho$ is directly proportional to change in temperature dT and original resistivity ρ_0 .

$$d\rho \propto \rho_0$$

$$d\rho \propto dT$$

Combining both eq

$$d\rho \propto \rho_0 dT$$

$$d\rho = \alpha \rho_0 dT$$

$$\alpha = \frac{d\rho}{\rho_0 dT}$$

Where α is proportionality constant called temperature coefficient of resistivity. It is fractional change in resistivity per unit change in temperature. Its SI unit is called K^{-1} . Its non-SI unit is called $^{\circ}C^{-1}$.

The final resistivity of material is given as

$$\text{Final resistivity} = \text{Original resistivity} + \text{Increase in resistivity}$$

$$\rho_t = \rho_0 + d\rho$$

$$\rho_t = \rho_0 + \alpha \rho_0 dT$$

$$\rho_t = \rho_0 (1 + \alpha dT)$$

The resistance and resistivity are related as $R = \frac{\rho L}{A}$. Using this relation we get

$$R_t = R_0 (1 + \alpha dT)$$

Q. 6 Define current density. State and explain Ohm's law and write its macroscopic form and microscopic form in terms of current density and electric field intensity?

CURRENT DENSITY

The current flowing through a conductor per unit area is called current density. It is a vector quantity. Its SI unit is called A/m^2 . Its formula is

$$J = \frac{I}{A}$$

OHM'S LAW

The Ohm's law can be stated in two ways called macroscopic form and microscopic form.

MACROSCOPIC FORM OF OHM'S LAW

The macroscopic form of Ohm's law states that current flows through a conductor when pd is applied at the ends of this conductor. The amount of this current is directly proportional to applied pd provided that physical state of conductor such as temperature remains constant.

Consider a conductor through which current I flows due to applied potential difference V , then by definition

$$V \propto I$$

$$V = IR$$

Where R is proportionality constant called resistance of conductor.

RESISTANCE

The measure of opposition to flow of charges through a conductor due to collision with ionic cores of conductor is called resistance. The ratio of pd applied at the end of a conductor and current flowing through conductor is equal to resistance.

The conductor used in a circuit to provide specified resistance is called resistor. Its symbol is in diagram.



UNIT

The SI unit of resistance is called ohm (Ω). The resistance of a conductor is called one ohm when one ampere current passes through a conductor due to one volt pd applied at the ends of a conductor.

OHMIC AND NON-OHMIC CIRCUITS

The material or circuit element which obeys Ohm's law is called ohmic. The graph between current and voltage is a straight line.

The material or circuit element such as diode, bulb which does not obey Ohm's law is called non-ohmic. The graph between current and voltage is non linear.

MICROSCOPIC FORM OF OHM'S LAW

The microscopic form of Ohm's law states that current density established in conductor is directly proportional to electric field which is applied at the ends of a conductor.

EXPLANATION

Consider a conductor having length L and cylindrical cross sectional area A . The current I flows through conductor when Pd ' V ' is applied at its end. The current is directly proportional to applied Pd.

$$V = IR$$

$$\text{As } R = \rho L/A$$

$$V = I \frac{\rho L}{A}$$

$$V = J \rho L$$

Where current per unit area is called current density $J = \frac{I}{A}$

$$\frac{V}{L} = J \rho$$

----- (1)

Potential difference per unit length (V/L) of conductor is equal to electric field E
 ----- (2)

$$\frac{V}{L} = E$$

Comparing eq(1) and eq(2)

$$E = J \rho$$

This is called microscopic form of Ohm's law.

RELATION BETWEEN MICROSCOPIC AND MACROSCOPIC PARAMETERS OF OHM'S LAW

The macroscopic parameter current I and microscopic parameter current density of Ohm's law are related as

$$I = \int \vec{J} \cdot d\vec{a}$$

The macroscopic parameter potential difference V and microscopic parameter electric field E of Ohm's law are related as

$$V = \int \vec{E} \cdot d\vec{r}$$

The macroscopic parameter resistance R and microscopic parameter resistivity ρ of Ohm's law are related as

$$R = \frac{\rho L}{A}$$

Q. 7 Describe how electrical energy is transferred to any other form of energy in electrical circuit. Also define electrical power and write its unit.

ENERGY TRANSFER IN ELECTRIC CIRCUIT

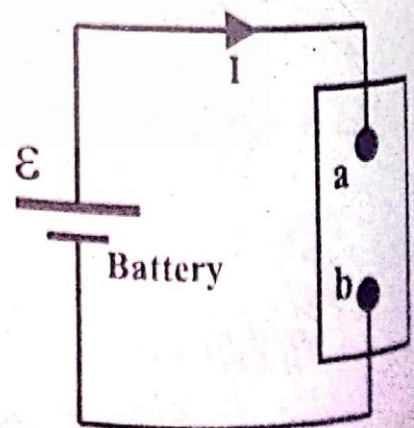
Consider an electric circuit in which a battery is connected with a black box as shown in fig. The steady current I is established in wires that causes steady potential difference V_{ab} between terminal "a" and "b". The box whose contents are unknown is called black box. It may have a resistor, a motor or a storage battery.

The terminal "a" has high potential and terminal "b" has low potential. The electrical potential energy of charge dq decreases by amount dqV_{ab} when charge dq moves from point a (high potential) to point b (low potential) inside the box.

The conservation of energy tells that this energy is transferred in the box from electrical energy to some other form of energy. The amount of electrical energy dU_e transferred inside the box in time dt is given as

$$dU_e = dq V_{ab}$$

$$dU_e = I dt V_{ab}$$



$$\frac{dU_e}{dt} = I V_{ab}$$

The term $\frac{dU_e}{dt}$ is rate of energy transfer called electric power

$$P_e = I V_{ab}$$

- ① The energy appears as mechanical work done by the motor if device in a black box is motor.
- ② The energy appears as stored chemical energy in this second battery if device is a storage battery in the black box.
- ③ The energy appears as internal energy if device in black box is a resistor.

POWER DISSIPATION

Now consider a special case that device inside box is a resistor. Then by Ohm's law

$$V_{ab} = I R$$

The electrical power dissipated in resistor is

$$P_e = I (I R)$$

$$P_e = I^2 R$$

Also we can write power as

$$P_e = \frac{V^2}{R^2} R = \frac{V^2}{R}$$

The eq $P_e = I^2 R$ and eq $P_e = V^2/R$ are called Joule's law. The conversion of electrical energy into heat energy in a resistor is called Joule heating effect.

UNIT

The unit of electrical power is called watt. The unit watt is equal to ampere – volt for electrical power.

The unit watt is equal to J/s for mechanical power. The bigger units of power commonly used are called kilowatt and megawatt.

The unit kilowatt-hour (kwh) is used for electrical energy.

$$\text{kwh} = 10^3 \left(\frac{\text{J}}{\text{s}} \right) 3600 \text{ s}$$

$$\text{kwh} = 3.6 \times 10^6 \text{ J} = 3.6 \text{ MJ}$$

Q. 8 Show that metals obey Ohm's law taken into account microscopic form of Ohm's law. Or derive a relation $\rho = m/ne^2\tau$.

METALS OBEY OHM'S LAW

The Ohm's law is not a fundamental law of electromagnetism. It depends upon properties of conducting material.

Almost all the metals are conductors and having conduction electrons and ionic cores. The assembly of conduction electrons is called electron gas. The conduction electrons move randomly and collide with ionic cores of the lattice in the absence of electric field. They having mean free path λ (average distance between collisions) and mean free time τ (average time between collisions).

The electrons modify their random motion when electric field is applied to metal in such a way that they drift slowly in the direction opposite to the direction of the applied electric field with average drift speed V_d .

The electric force F_e experienced by electron having charge e due to applied electric field E is

$$F_e = e E \quad \text{----- (1)}$$

This electric force F_e produces acceleration 'a' in electron having mass m

$$F_e = m a \quad \text{----- (2)}$$

Comparing eq (1) and eq(2)

$$m a = e E$$

$$a = \frac{e E}{m} \quad \text{----- (3)}$$

The acceleration in terms of drift velocity V_d and mean free time τ between collisions is given as

$$a = \frac{V_d}{\tau} \quad \text{----- (4)}$$

Comparing eq(3) and eq(4)

$$\frac{V_d}{\tau} = \frac{e E}{m}$$

$$V_d = \frac{e E \tau}{m} \quad \text{----- (5)}$$

The drift velocity in terms of current density is

$$V_d = \frac{J}{n e} \quad \text{----- (6)}$$

Comparing eq (5) and eq (6)

$$\frac{J}{n e} = \frac{e E \tau}{m}$$

$$J = \frac{n e^2 E \tau}{m} \quad \text{----- (7)}$$

The microscopic form of Ohm's law is

$$J = \frac{E}{\rho} \quad \text{----- (8)}$$

Comparing eq (7) and eq(8)

$$\frac{E}{\rho} = \frac{n e^2 E \tau}{m}$$

$$\rho = \frac{m}{n e^2 \tau}$$

The resistivity ρ of a metal will be constant if right side of above eq is constant. The parameters m is mass of electron, e is charge of electron and n is density of electrons. Therefore m , n and e are constant.

The parameter τ depends upon the speed distribution of conduction electrons and slightly affected by applied electric field so independent of applied electric field. Hence right side of equation is constant

It is therefore concluded that parameter ρ is constant for metals. The microscopic form of Ohm's law is

$$E = \rho J$$

When ρ is constant

$$E \propto J$$

Hence it is concluded that all metals obey Ohm's law.

Q. 9 Describe analogy between current and heat flow?

ANALOGY BETWEEN CURRENT AND HEAT FLOW

The flow of charges through a conductor due to potential difference and flow of heat through a conductor due to temperature difference has close analogy between them.

CURRENT FLOW IN CONDUCTOR

Consider a conducting slab having thickness dx and cross sectional area A . The current I flows through a conductor when potential difference dV is maintained at the ends of conductor.

$$I = \frac{V_a - V_b}{R} = - \frac{V_b - V_a}{R}$$

$$I = - \frac{dV}{R}$$

$$I = - \frac{A dV}{\rho dx} \quad [R = \rho L/A \text{ and } L = dx]$$

$$I = - A \sigma \frac{dV}{dx} \quad [\sigma = 1/\rho]$$

$$\frac{dq}{dt} = - A \sigma \frac{dV}{dx}$$

The negative sign means positive charge flows in the direction of decreasing V . It means dq/dt is positive when dV/dx is negative.

HEAT FLOW IN CONDUCTOR

Now consider same slab of homogeneous metallic material having length L . The cross sectional area of each of opposite faces of slab is A . The temperature of left face becomes T_1 when it is heated which is higher than temperature T_2 of colder face.

The heat Q flows from hotter face towards colder face. The rate of conduction of thermal energy H is defined as

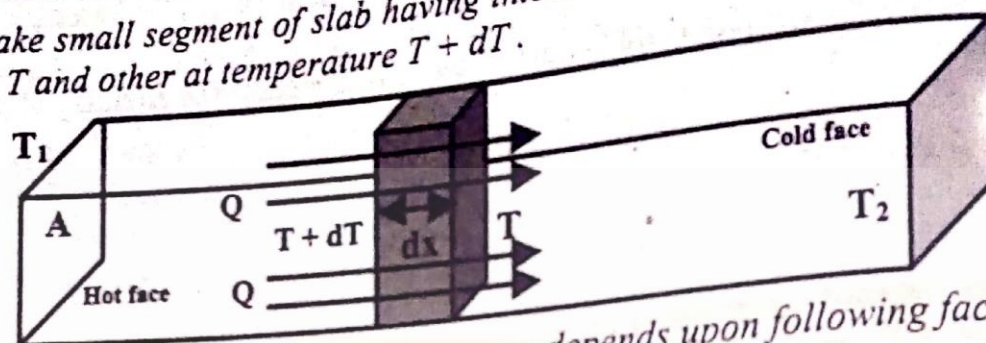
$$\text{Rate of conduction of thermal energy} = \frac{\text{Heat}}{\text{time}}$$

$$H = \frac{Q}{t}$$

The SI unit of rate of conduction of thermal energy is called watt.

FACTORS AFFECTING RATE OF CONDUCTION OF THERMAL ENERGY

Now take small segment of slab having thickness dx , face area A . Its one face is at temperature T and other at temperature $T + dT$.



The rate of conduction of thermal energy depends upon following factors.

1: CROSS-SECTION AREA OF THE MATERIAL

The larger cross-sectional area A of a solid material contains larger number of molecules and free electrons on each layer parallel to its cross-sectional area and hence greater will be rate of conduction of thermal energy through the material. Therefore rate of conduction of thermal energy is directly proportional to cross-sectional area of material.

$$H \propto A \quad \text{----- (1)}$$

2: LENGTH OF THE MATERIAL

The rate of conduction of thermal energy will be smaller when larger is the length between the hot and cold ends of the solid because it will take more time to conduct heat to the colder end. Hence rate of conduction of thermal energy is inversely proportional to length of material.

$$H \propto \frac{1}{dx} \quad \text{----- (2)}$$

3: TEMPERATURE DIFFERENCE BETWEEN ENDS OF MATERIAL

The rate of conduction of thermal energy is directly proportional to temperature difference. It means rate of conduction of thermal energy is large when temperature difference between opposite faces is large.

$$H \propto dT \quad \text{----- (3)}$$

These three experimental facts can be summed up as

$$H \propto A \frac{dT}{dx}$$

$$H = -k A \frac{dT}{dx}$$

The negative sign means heat flows in the direction of decreasing $\frac{dT}{dx}$. The term $\frac{dT}{dx}$ is called temperature gradient. Where k is constant of proportionality called thermal conductivity.

COMPARISON OF BOTH RELATIONS

The rate of flow of charges is given as

$$I = -A \sigma \frac{dV}{dx}$$

The rate of flow of thermal energy is

$$H = -k A \frac{dT}{dx}$$

The thermal conductivity k corresponds to electrical conductivity σ . The rate of charge flow (I) corresponds to rate of heat flow (H). The potential gradient (dV/dx) corresponds to temperature gradient (dT/dx).

It is concluded that both heat carrier and charge carrier in a conductor are electrons. The good conductors of heat are also good conductors of electricity.

Q.10 Write a note on superconductivity?**SUPER CONDUCTIVITY**

The resistivity of conductors decreases as we reduced the temperature because scattering of electrons by atoms vibrating from their equilibrium lattice positions decreases. The atoms retain certain minimum vibrational motion even at the absolute zero of temperature. Some materials show zero resistivity below certain temperature called critical temperature. These materials are called superconductors. The study of properties associated with these material is called superconductivity.

The first superconductor was discovered in 1911 by Kmaerlingh Ornes when resistivity of mercury becomes zero below temperature 4.2 K.

Some metals including aluminum at $T_c = 1.18$ K, tin at $T_c = 3.72$ K and lead at $T_c = 7.2$ K show superconductivity at these temperatures.

In 1986, a new class of ceramic materials was discovered that show superconductivity at $T_c = 125$ K.

Recently a complex crystalline structure called barium copper oxide shows superconductivity at 163 K by Yao Lian's, Lee. There is no superconductor discovered at room temperature.

There are many applications of superconductors in technology. The magnetic resonance imaging called MRI uses strong magnetic field produced by superconducting materials for scanning computer processing produces the image identifying tumors and inflamed tissues.

The superconductors are used in magnetic levitation trains, powerful but small electric motors and faster computer chips.

SHORT QUESTIONS

Q:1 Is filament resistance lower or higher in a 500 W light bulb than in a 100 W bulb when both are designed to operate at 220 V

Ans: The resistance of 500 W bulb operating at 220 V is

$$R_1 = \frac{V^2}{P} = \frac{(220)^2}{500} = 97 \Omega$$

The resistance of 100 W bulb operating at 220 V is

$$R_2 = \frac{V^2}{P} = \frac{(220)^2}{100} = 484 \Omega$$

Hence $R_1 < R_2$

Q:2 Why is it better to send 10 MW of electric power over long distances at 10 kV rather than 220 V.

Ans: The loss in power when 10 MW of electric power is sent over long distances at 10 kV.

$$P_{\text{loss}} = I^2 R = \left(\frac{P}{V}\right)^2 R = \left(\frac{10 \times 10^6}{10 \times 10^3}\right)^2 R = 10^{-6} R \text{ Watt}$$

The loss in power when 10 MW of electric power is sent over long distances at 220 V

$$P_{\text{loss}} = I^2 R = \left(\frac{P}{V}\right)^2 R = \left(\frac{10 \times 10^6}{220}\right)^2 R = 2 \times 10^{-9} R \text{ Watt}$$

Therefore, it is better to send 10 MW of electrical power at 10 kV because power losses are small.

Q:3 Discuss the difficulties of testing whether the filament of a lighted bulb obey Ohm's law.

Ans: The Ohm's law states that Pd across a conductor is directly proportional to current flowing through it provided that temperature is constant. When Ohm's law is tested with filament of a lighted bulb, temperature does not remain constant. Therefore, temperature is the difficulty because resistance changes.

Q:4 Using the formula for current density $J = n e V_d$, find the unit of J in SI system.

Ans:

$$J = n e V_d = \frac{\text{No. of electrons}(\text{Charge of electrons})(\text{Velocity})}{\text{Volume}}$$

$$J = \frac{(C)(m/s)}{m^3} = \frac{A}{m^2}$$

Q:5 What special characteristics must heating wire have

Ans: The heating wire must dissipate power at high rate. This is possible when wire has high resistance and small cross-sectional area.

Q:6 Name other physical quantities that like current are scalars having a sense represented by an arrow in a diagram?

Ans: The name of other physical quantity that like current is scalar having a sense represented by an arrow in a diagram is emf.

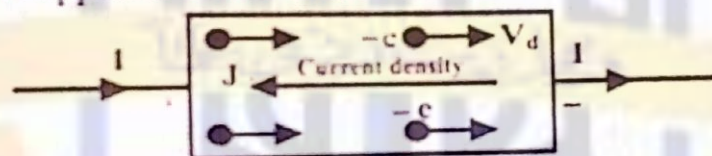
Q:7 In our convention for the direction of current arrows (a) would it have been more convenient or even possible, to assume all charge carries to be negative (b) would it have been more convenient, or even possible to have labeled electrons as positive, the proton as negative?

Ans: (a) Yes, it would be more convenient or even possible, to assume all charge carries to be negative

(b) Yes, it would be more convenient, or even possible to labeled electrons as positive and the proton as negative.

Q:8 A current I enters one corner of a square sheet of copper and leaves at the opposite corner. Sketch arrows at various points within the square to represent the relative values of the current density J .

Ans: The diagram in which current I enters into one corner of a square sheet of copper and leaves at the opposite corner is shown below



Q:9 A pd V is applied to a copper wire of diameter d and length L . What is the effect on the drift speed of (a) doubling V (b) doubling L and (c) doubling d ?

Ans: (a) The drift velocity becomes double when V is made double

$$V_d = \frac{J}{ne} = \frac{1}{ne} \left(\frac{I}{A} \right) = \frac{1}{neA} \left(\frac{V}{R} \right)$$

(b) The drift velocity becomes half when L is made double $\left[V_d = \frac{V}{ne} \left(\frac{1}{L \rho} \right) \right]$

(c) The drift velocity reduces four times when d is made double

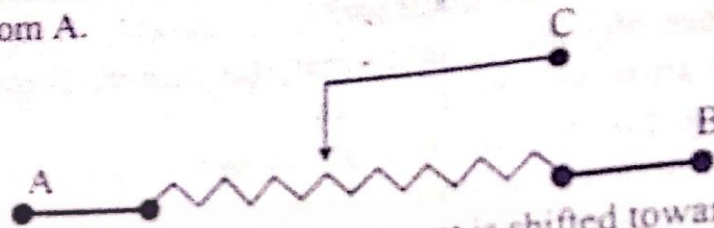
$$V_d = \frac{J}{ne} = \frac{1}{ne} \left(\frac{I}{A} \right) = \frac{I}{ne \pi r^2} = \frac{4I}{ne \pi d^2}$$

Q:10 Why is it not possible to measure the drift speed for electrons by timing their travel along a conductor?

Ans: The drift speed for electrons when traveling along a conductor is very small. It is of the order of few cm/s. The value of time must be very small. Therefore, it is not possible to measure the drift speed for electrons by timing their travel along a conductor.

Q:11 Describe briefly some possible designs of variable resistors

Ans: The rheostat acts as variable resistor as shown in fig. the fixed terminal say A and the sliding terminal C are inserted in the circuit. In this way the resistance of the wire between A and C is used. The resistance increases when sliding contact is shifted away from A.



The resistance decreases when sliding contact is shifted towards terminal A.

Q:12 Will the drift velocity of electrons in a current carrying metal conductor change when the temperature of the conductor is increased.

Ans: When temperature of a conductor is increased, its resistance increases. As a result, value of current decreases. The current and drift velocity are related as

$$V_d = \frac{J}{ne} = \frac{I}{neA}$$

It means current is directly proportional to drift velocity.

Therefore, drift velocity also decreases.

Q:13 How does the relation $V = IR$ apply to resistors that do not obey Ohm's law

Ans: The relation $V = IR$ to resistors that do not obey Ohm's law is written as $I = kV^n$, where k and n are constants.

Q:14 Why does an incandescent light bulb grow dimmer with use.

Ans: An incandescent light bulb grows dimmer with use because its resistance increases and its metallic character of filament changes.

Q:15 From a student's paper: The relationship $R = V/I$ tells us that resistance of a conductor is directly proportional to Pd . What do you think of this proportionality?

Ans: The resistance of conductor is a constant parameter. It will not change with variation in Pd . Hence relation $R \propto V$ is not correct. It must be $V \propto I$.

Q:16 Five wires of the same length and diameter are connected in turn between two points maintained at constant Pd . Will internal energy be developed at a fastest rate in the wire of (a) the smallest or (b) the largest resistance?

Ans: The internal energy will be developed at a faster rate in the wire having the smallest resistance.

Q:17 What type of temperature coefficient of resistivity is possessed by carbon and copper. Explain your answer.

Ans. The temperature coefficient of resistivity is given as

$$\alpha = \frac{d\rho}{\rho_0 dT}$$

The carbon has a negative temperature coefficient of resistivity because its resistivity drops as its temperature increases. The copper has a positive temperature coefficient of resistivity because its resistivity increases with increases in temperature.

SAMPLE PROBLEMS

- P.1 One end of Al wire whose diameter is 2.5 mm is welded to one end of Cu wire whose diameter is 1.8 mm. The composite wire carries a steady current of 1.3 A. What is current density in each wire?

Sol.

DATA

$$d = 2.5 \times 10^{-3} \text{ m}$$

$$I = 1.3 \text{ A}$$

$$J = ?$$

CALCULATIONS

The current density in Al is

$$J = \frac{I}{A} = \frac{I}{\pi r^2} = \frac{4I}{\pi d^2}$$

$$J = \frac{4 \times 1.3}{\pi \times (2.5 \times 10^{-3})^2} = 264968.15 \text{ A/m}^2$$

The current density in Cu wire is

DATA

$$d = 1.8 \times 10^{-3} \text{ m}$$

$$I = 1.3 \text{ A}$$

$$J = ?$$

$$J = \frac{4I}{\pi d^2}$$

$$J = \frac{4 \times 1.3}{\pi \times (1.8 \times 10^{-3})^2} = 511126.8 \text{ A/m}^2$$

- P.2 What is the drift speed of the conduction electrons in copper wire having molar mass $63.5 \times 10^{-3} \text{ kg/mol}$, density $8.96 \times 10^3 \text{ kg/m}^3$ and current density 511126.8 A/m^2 .

Sol.

DATA

$$\rho = 8.96 \times 10^3 \text{ kg/m}^3$$

$$M = 63.5 \times 10^{-3} \text{ kg/mol}$$

$$J = 511126.8 \text{ A/m}^2$$

$$N_A = 6.02 \times 10^{23} \text{ electrons/mol}$$

$$v_d = ?$$

CALCULATIONS

$$n = \frac{N_A \rho}{M}$$

$$n = \frac{6.02 \times 10^{23} \times 8.96 \times 10^3}{63.5 \times 10^{-3}} = 8.49 \times 10^{28}$$

Drift speed of conduction electrons

$$V_d = \frac{J}{n e}$$

$$V_d = \frac{511126.8}{8.49 \times 10^{28} \times 1.6 \times 10^{-19}} = 3.8 \times 10^{-5} \text{ m/s}$$

- P.3** A strip of silicon of width $W = 3.2 \text{ mm}$ and thickness $d = 250 \mu\text{m}$ carries a current of 190 mA . In this case $n = 8 \times 10^{21} \text{ electrons/m}^3$ (a) What is current density in strip (b) What is drift velocity?

Sol.

DATA

$$d = 250 \times 10^{-6} \text{ m}$$

$$W = 3.2 \times 10^{-3} \text{ m}$$

$$I = 190 \times 10^{-3} \text{ A}$$

$$J = ?$$

$$n = 8 \times 10^{21} \text{ electrons/m}^3$$

$$V_d = ?$$

CALCULATIONS

$$J = \frac{I}{A} = \frac{I}{W d}$$

$$J = \frac{190 \times 10^{-3}}{3.2 \times 10^{-3} \times 250 \times 10^{-6}} = 237500 \text{ A/m}^2$$

Drift velocity

$$V_d = \frac{J}{n e}$$

$$V_d = \frac{237500}{8 \times 10^{21} \times 1.6 \times 10^{-19}} = 185 \text{ m/s}$$

P.4

A rectangular block of iron has dimensions: $1.2 \text{ cm} \times 1.2 \text{ cm} \times 15 \text{ cm}$ (a) What is the resistance of the block measured between the two square ends (b) What is resistance between two opposite faces? The resistivity of iron at room temperature is $9.68 \times 10^{-8} \Omega\text{-m}$.

Sol.

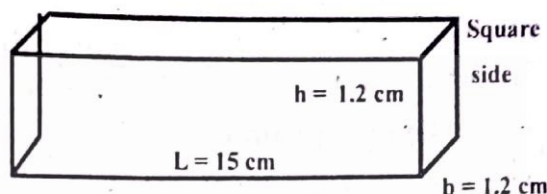
DATA

$$L = 15 \times 10^{-2} \text{ m}$$

$$b = 1.2 \times 10^{-2} \text{ m}$$

$$h = 1.2 \times 10^{-2} \text{ m}$$

$$\rho = 9.68 \times 10^{-8} \Omega\text{-m}$$



CALCULATIONS

Area of square ends

$$A = 1.2 \text{ cm} \times 1.2 \text{ cm} = 1.44 \times 10^{-4} \text{ m}^2$$

Resistance of block between square ends is

$$R = \frac{\rho L}{A}$$

$$R = \frac{9.68 \times 10^{-8} \times 15 \times 10^{-2}}{1.44 \times 10^{-4}} = 1 \times 10^{-4} \Omega$$

Area of rectangular side

$$A = 15 \times 1.2 \times 10^{-4} = 1.8 \times 10^{-3} \text{ m}^2$$

Resistance of block between opposite ends is

$$R = \frac{\rho L}{A}$$

$$R = \frac{9.68 \times 10^{-8} \times 1.2 \times 10^{-2}}{1.8 \times 10^{-3}} = 6.50 \times 10^{-7} \Omega$$

P.5

What is the mean free time between collisions for the conduction electrons in Cu? (b) What is mean free path (λ) for these collisions? Assume an effective speed V is $1.6 \times 10^6 \text{ m/s}$.

Sol.

DATA

$$m = 9.1 \times 10^{-31} \text{ kg}$$

$$\rho = 1.77 \times 10^{-8} \Omega\text{-m}$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

$$n = 8.49 \times 10^{28} \text{ electrons/m}^3$$

$$\langle V \rangle = 1.6 \times 10^6 \text{ m/s}$$

CALCULATIONS

$$\tau = \frac{m}{n e^2 \rho}$$

$$\tau = \frac{9.1 \times 10^{-31}}{8.49 \times 10^{28} \times (1.6 \times 10^{-19})^2 \times 1.77 \times 10^{-8}}$$

$$\tau = 2.48 \times 10^{-14} \text{ s}$$

The mean path is

$$\lambda = \tau \langle V \rangle$$

$$\lambda = 2.48 \times 10^{-14} \times 1.6 \times 10^6 = 4.0 \times 10^{-8} \text{ m}$$

- P.6 You are given a length of heating wire made of nickel-chromium-iron alloy called Nichrome. It has a resistance of 72Ω . It is to be connected across a 120 V line. Under what circumstance wire will dissipate more heat (a) Its entire length is connected across line (b) the wire is cut into half and two halves are connected in parallel across the line?

Sol.

DATA

$$V = 120 \text{ V}$$

$$R = 72 \Omega$$

$$P = ? \text{ (For entire wire)}$$

$$P = ? \text{ (For half wire)}$$

CALCULATIONS

$$(a) \quad P = \frac{V^2}{R} = \frac{(120)^2}{72} = 200 \text{ W}$$

$$(b) \quad P = \frac{V^2}{R/2} = \frac{(120)^2}{36} = 400 \text{ W}$$

Power for both halves of wire is 800 W . It is 4 times that for single wire

EXERCISE PROBLEMS

- P.1 An aluminum wire of certain diameter carries a steady current 3.14 A. If current density is 25 A/cm^2 , What should be the radius of the wire.

Sol.

DATA

$$I = 3.14 \text{ A}$$

$$J = 25 \text{ A/cm}^2 = 25 \times 10^4 \text{ A/m}^2$$

$$r = ?$$

CALCULATIONS

$$J = \frac{I}{A} = \frac{I}{\pi r^2}$$

$$r = \sqrt{\frac{I}{\pi J}}$$

$$r = \sqrt{\frac{3.14}{3.14 \times 25 \times 10^4}} = 2.0 \times 10^{-4} \text{ m}$$

- P.2 A strip of silicon carries current such that current density in the strip is 24 A/cm^2 . After doping the number of charge carriers (electrons) per unit volume are raised upto $8 \times 10^{21} \text{ m}^{-3}$. What is the drift velocity of charge carriers.

Sol.

DATA

$$J = 24 \text{ A/cm}^2 = 24 \times 10^4 \text{ A/m}^2$$

$$n = 8 \times 10^{21} \text{ m}^{-3}$$

$$V_d = ?$$

CALCULATIONS

$$V_d = \frac{J}{n e}$$

$$V_d = \frac{24 \times 10^4}{8 \times 10^{21} \times 1.6 \times 10^{-19}}$$

$$V_d = 188 \text{ m/s}$$

P.3. You are given a nichrome wire of resistance 144Ω . The wire is cut into half and the two halves are connected in parallel across 240 V line. Calculate the power dissipated.

Sol.

DATA

$$R = 144 \Omega$$

$$V = 240 \text{ V}$$

$$P = ?$$

$$R_1 = 72 \Omega$$

$$R_2 = 72 \Omega$$

CALCULATIONS

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2} = \frac{72 \times 72}{72 + 72} = 36 \Omega$$

Power dissipated

$$P = \frac{V^2}{R_{eq}} = \frac{(240)^2}{36} = 1600 \text{ W}$$

P.4 A current of 4.82 A exists in a 12.4Ω resistor for 4.6 minute. (a) How much charge passes (b) How many electrons pass through any cross section for this time?

Sol.

DATA

$$I = 4.82 \text{ A}$$

$$R = 12.4 \Omega$$

$$t = 4.6 \text{ min} = 4.6 \times 60 = 276 \text{ s}$$

$$q = ?$$

$$n = ?$$

CALCULATIONS.

(a)

$$q = I \times t$$

$$q = 4.82 \times 276 = 1330.32 \text{ C}$$

(b)

$$q = n e$$

$$n = \frac{q}{e}$$

$$n = \frac{1330.32}{1.6 \times 10^{-19}} = 8.31 \times 10^{21} \text{ electrons}$$

- P.5 The current in electron beam of a typical video display terminal is $200 \mu\text{A}$. How many electrons strike the screen each minute?

Sol.

DATA

$$I = 200 \times 10^{-6} \text{ A}$$

$$t = 1 \text{ min} = 60 \text{ s}$$

$$n = ?$$

CALCULATIONS

$$q = I \times t = 200 \times 10^{-6} \times 60 = 0.012 \text{ C}$$

The amount of electron

$$n = \frac{q}{e}$$

$$n = \frac{0.012}{1.6 \times 10^{-19}} = 7.5 \times 10^{16} \text{ electron}$$

- P.6 Suppose that we have 2.10×10^8 doubly charged positive ions per cubic cm, all moving with a speed of $1.4 \times 10^5 \text{ m/s}$. Calculate the current density in magnitude and direction. ($e = 1.6 \times 10^{-19} \text{ C}$)

Sol.

DATA

Double density is $2n$ and given as

$$2n = 2 \times 2.1 \times 10^8 \text{ ion/cm}^3 = 2 \times 2.1 \times 10^8 \times 10^6 \text{ ion/m}^3$$

$$2n = 4.2 \times 10^{14} \text{ ion/m}^3$$

$$V_d = 1.4 \times 10^5 \text{ m/s}$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

$$J = ?$$

CALCULATIONS

$$J = 2neV_d$$

$$J = 4.2 \times 10^{14} \times 1.6 \times 10^{-19} \times 1.4 \times 10^5$$

$$J = 9.4 \times 10^{-6} \text{ A/m}^2$$

- P.7 A small current of $123 \mu\text{A}$ exists in a Cu wire whose diameter is 2.46 mm . Calculate the current density and drift velocity of electrons?

Sol.

DATA

$$d = 2.46 \times 10^{-3} \text{ m}$$

$$I = 123 \times 10^{-12} \text{ A}$$

$$J = ?$$

$$V_d = ?$$

CALCULATIONS

$$J = \frac{I}{A} = \frac{I}{\pi r^2} = \frac{4I}{\pi d^2}$$

$$J = \frac{4 \times 123 \times 10^{-12}}{\pi (2.46 \times 10^{-3})^2}$$

$$J = 2.58 \times 10^{-5} \text{ A/m}^2$$

For drift velocity of conduction electron in Cu wire calculate value of n as

$$N_A = 6.02 \times 10^{23} \text{ electrons/mol}$$

$$\rho = 8.96 \times 10^3 \text{ kg/m}^3$$

$$M = 63.5 \times 10^{-3} \text{ kg/mol}$$

$$n = \frac{N_A \rho}{M} = \frac{6.02 \times 10^{23} \times 8.96 \times 10^3}{63.5 \times 10^{-3}} = 8.4 \times 10^{28}$$

Drift velocity of conduction electron in Cu wire

$$V_d = \frac{J}{ne}$$

$$V_d = \frac{2.58 \times 10^{-5}}{8.49 \times 10^{28} \times 1.6 \times 10^{-19}} = 1.89 \times 10^{-15} \text{ m/s}$$

- P.8** Suppose that a material composing a fuse melts and the current density rises 440 A/cm^2 . What diameter of cylindrical wire should be used for the fuse to limit the current to 0.552 A .

Sol.

DATA

$$d = ?$$

$$J = 440 \text{ A/cm}^2 = 440 \times 10^4 \text{ A/m}^2$$

$$I = 0.552 \text{ A}$$

CALCULATIONS

$$J = \frac{I}{A} = \frac{I}{\pi r^2} = \frac{4I}{\pi d^2}$$

$$d = \sqrt{\frac{4I}{\pi J}}$$

$$d = \sqrt{\frac{4 \times 0.552}{\pi \times 440 \times 10^4}} = 3.99 \times 10^{-4} \text{ m}$$

- P.9** A current is established in a gas-discharge tube when a sufficiently high Pd is applied across the two electrodes in the tube. The gas ionizes: electrons move towards the +ve terminal and singly charged +ve ions towards -ve terminal. What are the magnitude and direction of the current in a hydrogen discharged tube in which 3.1×10^{18} electrons and 1.1×10^{18} protons move apart a cross sectional area of tube each second.

Sol.

DATA

$$\text{No. of electrons} = 3.1 \times 10^{18}$$

$$\text{No. of protons} = 1.1 \times 10^{18}$$

$$q = 1.6 \times 10^{-19} (3.1 \times 10^{18} + 1.1 \times 10^{18})$$

$$q = 0.67 \text{ C}$$

$$t = 1 \text{ s}$$

CALCULATIONS

$$I = \frac{q}{t}$$

$$I = \frac{0.67}{1} = 0.67 \text{ A (towards negative terminal)}$$

- P.10 A PN junction is formed from two different semi-conducting materials in the form of identical cylinder with radius 0.165 mm. In one application 3.5×10^{15} electrons/s flow across the junction from n to P side while 2.25×10^{15} holes/s flow across the junction from P to n side. Find the total current and current density?

Sol.

DATA

$$I_e = 3.5 \times 10^{15} \text{ electrons/s} = 3.5 \times 10^{15} \times 1.6 \times 10^{-19}$$

$$I_e = 5.6 \times 10^{-4} \text{ A}$$

$$I_p = 2.25 \times 10^{15} \text{ holes/s} = 2.25 \times 10^{15} \times 1.6 \times 10^{-19} \text{ A}$$

$$I_p = 3.6 \times 10^{-4} \text{ A}$$

$$J = ?$$

$$I = ?$$

CALCULATIONS

$$I = I_e + I_p$$

$$I = (5.6 + 3.6) \times 10^{-4} = 9.2 \times 10^{-4} \text{ A}$$

Current density

$$J = \frac{I}{A} = \frac{I}{\pi r^2}$$

$$J = \frac{9.2 \times 10^{-4}}{\pi (0.165 \times 10^{-3})^2} = 10761.9 \text{ A/m}^2$$

- P.11 The belt of an electrostatic accelerator is 52 cm and travel at 28 m/s. The belt carries a charge into sphere at the rate corresponding to 95 μA . Compute the surface charge density on the belt?

Sol.

DATA

$$W = 52 \text{ cm}$$

$$V_d = 28 \text{ m/s}$$

$$I = 95 \mu\text{A}$$

$$\sigma = ?$$

CALCULATIONS

$$\sigma = \frac{q}{A} = \frac{1 \times 1}{A} = \frac{1 \times 1}{V_d \times 1 \times W}$$

$$\sigma = \frac{1}{V_d \times W}$$

$$\sigma = \frac{95 \times 10^{-6}}{28 \times 52 \times 10^{-2}} = 6.5 \times 10^{-6} \text{ C/m}^2$$

P.12 Near the earth, the density of protons in the solar wind is 8.70 per cm³ and their speed 470 km/s (a) Find the current density of these protons (b) If the earth magnetic field did not deflect them, the protons would strike the earth. What total current would the earth receive?

Sol.

DATA

$$R_e = 6400 \text{ km}$$

$$n_e = 8.70 \text{ proton cm}^{-3} = 8.70 \times 10^6 \times 1.6 \times 10^{-19} \text{ C/m}^3$$

$$V_d = 470 \text{ km/s}$$

$$J = ?$$

$$I = ?$$

CALCULATIONS

(a)

$$J = neV_d$$

$$J = 8.7 \times 10^6 \times 1.6 \times 10^{-19} \times 470 \times 10^3$$

$$J = 6.5 \times 10^{-7} \text{ A/m}^2$$

(b)

$$1 \text{ m}^2 \text{ cross section of earth receives current} = 6.5 \times 10^{-7} \text{ A}$$

$$\pi R_e^2 \text{ cross section of earth receives current} = 6.5 \times 10^{-7} \pi (6.4 \times 10^6)^2$$

$$I = 83.5 \times 10^6 \text{ A}$$

How long does it take electrons to get from a car battery to the starting motor. Assume that the current is 115 A and the electrons travel through copper wire with cross sectional area 31.2 mm² and length 85.5 cm.

Sol.

DATA

$$I = 115 \text{ A}$$

$$A = 31.2 \text{ mm}^2$$

$$L = 85.5 \text{ cm}$$

$$t = ?$$

CALCULATIONS

$$J = \frac{I}{A}$$

$$J = \frac{115}{31.2 \times 10^{-6}} = 3.6 \times 10^6 \text{ A/m}^2$$

$$n = 8.49 \times 10^{28} \text{ for Cu wire}$$

$$V_d = \frac{J}{ne} = \frac{3.6 \times 10^6}{8.49 \times 10^{28} \times 1.6 \times 10^{-19}}$$

$$V_d = 2.71 \times 10^{-4} \text{ m/s}$$

$$2.71 \times 10^{-4} \text{ m distance covered in time} = 1 \text{ sec}$$

$$1 \text{ m distance covered in time} = \frac{1}{2.71 \times 10^{-4}}$$

$$85.5 \times 10^{-2} \text{ m distance covered in time} = \frac{1}{2.71 \times 10^{-4}} \times 85.5 \times 10^{-2}$$

$$t = 3151 \text{ sec} = 52.5 \text{ min}$$

P.14 A steel trolley car rail has a cross sectional area of 56 cm^2 . What is the resistance of 11 km of rail. The resistivity of steel is $3 \times 10^{-7} \Omega \cdot \text{m}$

Sol.

DATA

$$A = 56 \text{ cm}^2$$

$$L = 11 \text{ km}$$

$$\rho = 3 \times 10^{-7} \Omega \cdot \text{m}$$

$$R = ?$$

CALCULATIONS

$$R = \frac{\rho L}{A}$$

$$R = \frac{3 \times 10^{-7} \times 11 \times 10^3}{56 \times 10^{-4}} = 0.58 \Omega$$

P.15 A wire 4m long and 6 mm in diameter has a resistance of $15 \text{ m}\Omega$. A potential difference of 23V is applied between the ends. What is current, current density and resistivity.

Sol.

DATA

$$L = 4 \text{ m}$$

$$d = 6 \text{ mm} = 6 \times 10^{-3} \text{ m}$$

$$R = 15 \times 10^{-3} \Omega$$

$$V = 23 \text{ Volt}$$

CALCULATIONS

$$I = \frac{V}{R}$$

$$I = \frac{23}{15 \times 10^{-3}} = 1533.3 \text{ A}$$

$$J = \frac{I}{A} = \frac{I}{\pi r^2} = \frac{4I}{\pi d^2}$$

$$J = \frac{4 \times 1533.3}{\pi (6 \times 10^{-3})^2} = 54256900 \text{ A/m}^2$$

Resistivity is

$$R = \frac{\rho L}{A} = \frac{\rho L J}{I}$$

$$\rho = \frac{I R}{L J} = \frac{1533.3 \times 15 \times 10^{-3}}{4 \times 54256900}$$

$$\rho = 1.05 \times 10^{-7} \Omega\text{-m}$$

P.16 A fluid with resistivity $9.40 \Omega\text{-m}$ seeps into the space between the plates of a 110 pF parallel plate air capacitor. When the space is completely filled with the fluid, what is the resistance between plates?

Sol.

DATA

$$\rho = 9.40 \Omega\text{-m}$$

$$C = 110 \times 10^{-12} \text{ F}$$

$$R = ?$$

CALCULATIONS

$$R = \frac{\rho L}{A}$$

But capacitance of parallel plate capacitor is $C = \frac{A \epsilon_0}{L}$. Using both eq

$$R = \frac{\rho \epsilon_0}{C}$$

$$R = \frac{9.40 \times 8.85 \times 10^{-12}}{110 \times 10^{-12}} = 0.75 \Omega$$

17 At what temperature would the resistance of a copper conductor be double its resistance at 20°C ?

Sol.

DATA

$$T_2 = ?$$

$$T_1 = 20^\circ\text{C}$$

$$\alpha = 4.3 \times 10^{-3} ^\circ\text{C}^{-1}$$

$$\text{Resistance at } 20^\circ\text{C} = R_0$$

$$\text{Resistance at } T_2 = 2 R_0$$

CALCULATIONS

$$R_T = 2 R_0$$

$$R_0[1 + \alpha(T_2 - T_1)] = 2 R_0$$

$$[1 + \alpha(T_2 - T_1)] = 2$$

$$4.3 \times 10^{-3} (T_2 - 20) = 1$$

$$T_2 = 252.2^\circ\text{C}$$

- P.18 The copper windings of a motor have a resistance of $50\ \Omega$ at 20°C when the motor is idle. After running for several hours the resistance rises to $58\ \Omega$. What is the temperature of the windings of a motor?

Sol.

DATA

$$T_2 = ?$$

$$R_0 = 50\ \Omega$$

$$T_1 = 20^\circ\text{C}$$

$$R_T = 58\ \Omega$$

$$\alpha = 4.3 \times 10^{-3}^\circ\text{C}^{-1}$$

CALCULATIONS

$$R_T = R_0[1 + \alpha(T_2 - T_1)]$$

$$58 = 50[1 + 4.3 \times 10^{-3}(T_2 - 20)]$$

$$T_2 = 57^\circ\text{C}$$

- P.19 A wire with a resistance of $6\ \Omega$ is drawn out through a die so that its new length is three times its original length. Find the resistance of longer wire.

Sol.

DATA

$$R = 6\ \Omega$$

$$L' = 3L$$

$$A' = \frac{A}{3}$$

$$R' = ?$$

CALCULATIONS

$$R = \frac{\rho L}{A}$$

New resistance is

$$R' = \frac{\rho L'}{A'} = \frac{\rho(3L)}{A/3} = 9\left(\frac{\rho L}{A}\right)$$

$$R' = 9R = 9 \times 6$$

$$R' = 54\ \Omega$$

P.20 What must be the diameter of an iron wire if it is to have the same resistance as a copper wire 1.19 mm in diameter both wires have same length?

Sol.

DATA

$$d_{Fe} = ?$$

$$d_{Cu} = 1.19 \text{ mm}$$

$$R_{Fe} = R_{Cu}$$

$$L_{Fe} = L_{Cu} = L$$

$$\rho_{Fe} = 9.68 \times 10^{-8} \Omega\text{-m}$$

$$\rho_{Cu} = 1.77 \times 10^{-8} \Omega\text{-m}$$

CALCULATIONS

$$R_{Fe} = R_{Cu}$$

$$\frac{\rho_{Fe} L}{A_{Fe}} = \frac{\rho_{Cu} L}{A_{Cu}}$$

$$\frac{\rho_{Fe}}{\pi d_{Fe}^2} = \frac{\rho_{Cu}}{\pi d_{Cu}^2}$$

$$d_{Fe} = \sqrt{\frac{\rho_{Fe} d_{Cu}^2}{\rho_{Cu}}}$$

$$d_{Fe} = \sqrt{\frac{9.68 \times 10^{-8} (1.19 \times 10^{-3})^2}{1.77 \times 10^{-8}}}$$

$$d_{Fe} = 2.78 \times 10^{-3} \text{ m}$$

P.21 When 115 V is applied across a 9.66 m long wire the current density is 1.42 A/cm^2 . Calculate conductivity of wire material?

Sol.

DATA

$$V = 115 \text{ V}$$

$$L = 9.66 \text{ m}$$

$$J = 1.42 \times 10^{-4} \text{ A/m}^2$$

$$\sigma = ?$$

CALCULATIONS

$$\sigma = \frac{L}{R A}$$

$$\sigma = \frac{J L}{V} \quad [V = I R \text{ and } J = \frac{I}{A}]$$

$$\sigma = \frac{1.42 \times 10^{-4} \times 9.66}{115} = 1192.8 (\Omega\text{-m})^{-1}$$

- P.22 A block in the shape of a rectangular solid has a cross sectional area of 3.50 cm^2 , length 15.8 cm , and resistance 935Ω . The material of block has 5.33×10^{22} conduction electrons/ m^3 . A Pd of 35.8 V is maintained between its ends (a) Find the current in the block (b) current density (c) drift velocity (d) electric field.

Sol.

DATA

$$A = 3.50 \times 10^{-4} \text{ m}^2$$

$$L = 15.8 \times 10^{-2} \text{ m}$$

$$R = 935 \Omega$$

$$n = 5.33 \times 10^{22} \text{ electron/m}^3$$

$$V = 35.8 \text{ Volt}$$

CALCULATIONS

$$(a) \quad I = \frac{V}{R}$$

$$I = \frac{35.8}{935} = 0.038 \text{ A}$$

$$(b) \quad J = \frac{I}{A}$$

$$J = \frac{0.038}{3.50 \times 10^{-4}} = 109 \text{ A/m}^2$$

$$(c) \quad V_d = \frac{J}{n e}$$

$$V_d = \frac{109}{5.33 \times 10^{22} \times 1.6 \times 10^{-19}} = 0.012 \text{ m/s}$$

$$(d) \quad E = \frac{V}{L}$$

$$E = \frac{35.8}{15.8 \times 10^{-2}} = 226.5 \text{ V/m}$$

- P.23 When a metal is heated not only its resistance but also its length and its cross sectional area change (a) if temperature changes by 1°C what fraction change in R , L and A occur from a copper conductor (b) What conclusion do you draw. $\alpha = 1.7 \times 10^{-5} \text{ C}^{-1}$

Sol.

DATA

$$dT = 1^\circ\text{C}$$

$$\alpha = 1.7 \times 10^{-5} \text{ C}^{-1}$$

$$\frac{R_1}{R_0} = ?$$

CALCULATIONS

$$R_T = R_0(1 + \alpha dT)$$

$$\frac{R_T}{R_0} = (1 + \alpha dT)$$

$$\frac{R_T}{R_0} = (1 + 1.7 \times 10^{-5})$$

$$\frac{R_T}{R_0} = 2.7 \times 10^{-5}$$

----- (1)

Now consider

$$\frac{R_T}{R_0} = \frac{\rho L_T}{A_T} \times \frac{A_0}{\rho L_0}$$

----- (2)

Comparing eq(1) and eq(2)

$$\frac{A_0 L_T}{A_T L_0} = 2.75 \times 10^{-5}$$

P.24 A student's 9V and 7.5W portable radio was left on from 9.00 pm until 3:00 am. How much charge passed through the wires?

Sol.

DATA

$$V = 9 \text{ Volt}$$

$$P = 7.5 \text{ W}$$

$$t = 6 \text{ h}$$

CALCULATIONS

$$P = IV$$

$$I = \frac{P}{V}$$

$$I = \frac{7.5}{9} = 0.83 \text{ A}$$

Charge passing through wire

$$q = I \times t$$

$$q = 0.83 \times 6 \times 3600 = 17928 \text{ C}$$

P.25 A space heater operating from a 120V line has a hot resistance of 14Ω (a) At what rate is electrical energy transferred into internal energy? (b) At 5.22 cent/kwh, what does it cost to operate the device for 6h 25 minutes?

Sol.

DATA

$$V = 120 \text{ Volt}$$

$$R = 14 \Omega$$

$$P = ?$$

CALCULATIONS

$$P = \frac{V^2}{R} = \frac{(120)^2}{14} = 1028.4 \text{ watt}$$

$$\text{Cost of one kwh} = 5.22 \text{ cent}$$

$$t = 6 \text{ h } 25 \text{ min} = 6 \times 3600 + 25 \times 60 = 23100 \text{ s}$$

Electrical Energy

$$E = P \times t$$

$$E = 1028.4 \times 23100 = 23756040 \text{ J}$$

$$E = \frac{23756040}{3.6 \times 10^6} = 6.6 \text{ kwh}$$

$$\text{Cost of 1 kwh of energy} = 5.22 \text{ cent}$$

$$\text{Cost of 6.5 kwh of energy} = 5.22 \times 6.6 = 34.4 \text{ cent}$$

P.26 A cylindrical resistor of radius 5.12 mm and length 1.96 cm is made of material that has resistivity of $3.5 \times 10^{-5} \Omega - \text{m}$ what are (a) current density (b) Pd when the power dissipation is 1.55 W.

Sol.

DATA

$$r = 5.12 \times 10^{-3} \text{ m}$$

$$L = 1.96 \times 10^{-2} \text{ m}$$

$$\rho = 3.5 \times 10^{-5} \Omega - \text{m}$$

$$P = 1.55 \text{ W}$$

$$J = ?$$

$$V = ?$$

CALCULATIONS

$$P = \frac{V^2}{R} = \frac{A V^2}{\rho L} = \frac{\pi r^2 V^2}{\rho L}$$

$$V = \sqrt{\frac{\rho L P}{\pi r^2}}$$

$$V = \sqrt{\frac{3.5 \times 10^{-5} \times 1.96 \times 10^{-2} \times 1.55}{\pi (5.12 \times 10^{-3})^2}}$$

$$V = 0.113 \text{ V}$$

Current is given as

$$I = \frac{P}{V} = \frac{1.55}{0.113} = 13.7 \text{ A}$$

Current density

$$J = \frac{I}{A} = \frac{13.6}{\pi (5.12 \times 10^{-3})^2}$$

$$J = 165222.5 \text{ A/m}^2$$

P.27 A coil of current carrying Nichrome wire is immersed in a liquid contained in a calorimeter. When Pd across the coil is 12V and the current through the coil is 5.2 A, the liquid boils at the steady rate, evaporating at the rate of 21 mg/s. Calculate the heat of vaporization of the liquid.

Sol.

DATA

$$L_v = ?$$

$$V = 12 \text{ Volt}$$

$$I = 5.2 \text{ A}$$

$$\frac{m}{t} = 21 \text{ mg/s} = 21 \times 10^{-3} \times 10^{-3}$$

CALCULATIONS

$$P = \frac{W}{t} = \frac{m L_v}{t} \quad \dots (1)$$

$$P = 1 \text{ W} \quad \dots (2)$$

On comparing both eq

$$L_v = \frac{5.2 \times 12}{21 \times 10^{-3}} = 2.97 \times 10^6 \text{ J/kg}$$

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